

LP Course Notes

Stephen Arndt

Instructor: Fatma Kılınç-Karzan

Fall 2024

Contents

1	Linear Algebra Background	2
1.1	Definitions about Sets	2
1.2	Matrices and Linear Mappings	3
1.3	Inner Products and Orthogonality	3
1.4	Affine Subspaces	4
2	Start to LP Duality	4
2.1	Basics	4
2.2	Polyhedral Representations	5
2.3	Linear, Affine, Conic, Convex	6
2.4	Classical Theorems	6
2.5	Farkas' Lemma and Certificates	8
2.6	LP Duality	8
3	Polyhedral Theory	10
3.1	Faces	10
3.2	Extreme Points	10
3.3	Cones	11
3.4	Extreme Rays	12
3.5	Bases	12
3.6	Interiors	12
3.7	General Representation Theorem	13
4	Simplex Algorithm	13
4.1	Algorithm	13
4.2	Definitions	13
4.3	Correctness (Non-Degeneracy)	14
4.4	Dealing with Degeneracy	15
4.5	Extensions (Algebraic Sensitivity)	15

5	Final Topics	15
5.1	Column Generation	16
5.2	Dantzig-Wolfe Decomposition	16
5.3	Bender's Decomposition	17

1 Linear Algebra Background

1.1 Definitions about Sets

Definition 1.1. A set $S \subseteq \mathbb{R}^n$ is **open** if for all $x \in S$, there is a ball $B_\epsilon(x) \subseteq S$ for some $\epsilon > 0$.

Definition 1.2. A set $S \subseteq \mathbb{R}^n$ is **closed** if $\text{cl}(S) = S$. The **closure** of a set S is S along with all of its limit points. A **limit point** x of S is a point (not necessarily in S) where all balls $B_\epsilon(x)$ for $\epsilon > 0$ intersect S .

Theorem 1.3. A set $S \subseteq \mathbb{R}^n$ is **compact** if it is closed and bounded.

In $\mathbb{R}^n$ the only sets which are “clopen” are $\emptyset$ and $\mathbb{R}^n$. Note that the intersection of a finite number of open sets is open and the intersection of any number of closed sets is closed.

Definition 1.4. The **interior** $\text{int}(S)$ of a set $S \subseteq \mathbb{R}^n$ is the largest open set contained in S .

Definition 1.5. The **relative interior** $\text{relint}(S)$ of a set $S \subseteq \mathbb{R}^n$ is the set of $x \in S$ where $B_\epsilon(x) \cap \text{aff}(S) \subseteq S$ for some $\epsilon > 0$.

For example, a convex 2D set in 3D space will have empty interior, but a 2D relative interior.

Definition 1.6. The **boundary** $\text{bd}(S)$ of a set $S \subseteq \mathbb{R}^n$ is $\text{cl}(S) - \text{int}(S)$.

Definition 1.7. The **relative boundary** $\text{rbd}(S)$ of a set $S \subseteq \mathbb{R}^n$ is $\text{cl}(S) - \text{relint}(S)$.

For example, a convex 2D set in 3D space will equal its boundary, but not its relative boundary.

Theorem 1.8 (Extreme Value Theorem). *If $f : C \rightarrow \mathbb{R}$ is a continuous function on a compact set $C \subseteq \mathbb{R}^n$, then f attains a minimum and maximum on C .*

An application of the above theorem is that if f is a continuous and strictly positive function on compact set $C \subseteq \mathbb{R}^n$, then f is lower-bounded by some $m > 0$.

1.2 Matrices and Linear Mappings

Definition 1.9. A **linear subspace** $L \subseteq \mathbb{R}^n$ is a nonempty set which is closed under addition and scalar multiplication. It can be represented as $L = \{x \in \mathbb{R}^n : Ax = 0\}$.

Theorem 1.10. If $L \subseteq \mathbb{R}^n$ is a linear subspace, then L has a **basis**, and all bases have the same number of vectors, which is called the **dimension** of L .

Definition 1.11. A **linear mapping** $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a function satisfying $A(x + y) = A(x) + A(y)$ and $A(\lambda x) = \lambda A(x)$.

Theorem 1.12. Every linear mapping can be represented as a matrix and every matrix gives a linear mapping.

Matrices are linear mappings! You can think of Ax as a linear combination of the columns of A where x gives the scalar values.

Definition 1.13. The determinant $\det(A)$ of a square matrix A is the volume of the parallelotope given by $Ae_1, Ae_2, \dots, Ae_n$.

Theorem 1.14. Let A be an $n \times n$ matrix. Then for any row i , $\det(A) = \sum_{j=1}^n a_{ij} C^{ij}$ where $C^{ij} = (-1)^{i+j} \det(A_{ij})$ and $A_{ij} = A$ with row i , column j eliminated.

Definition 1.15. Let A be an $m \times n$ matrix (mapping $\mathbb{R}^n \rightarrow \mathbb{R}^m$).

- $\text{Im}(A) \subseteq \mathbb{R}^m$ is the image of A and $\text{rank}(A) = \dim(\text{Im}(A))$.
- $\text{Ker}(A) \subseteq \mathbb{R}^n$ is the set of $x \in \mathbb{R}^n$ with $Ax = 0$ and $\text{nullity}(A) = \dim(\text{Ker}(A))$.

Theorem 1.16 (Rank-Nullity Theorem). $\text{rank}(A) + \text{nullity}(A) = n$.

1.3 Inner Products and Orthogonality

Definition 1.17. A **Euclidean inner product** is an operation $\langle \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ s.t. $\langle \rangle$ is linear w.r.t. x when y is fixed, $\langle x, y \rangle = \langle y, x \rangle$, and $\langle x, x \rangle \geq 0$ with equality iff $x = 0$.

Definition 1.18. The **standard EIP** is $\langle x, y \rangle = x^T y$.

Theorem 1.19. For any EIP, every linear function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is $\langle f, x \rangle$ for some $f \in \mathbb{R}^n$, and $\langle f, x \rangle$ is a linear mapping for every $f \in \mathbb{R}^n$.

Linear functions are inner products with a fixed vector – for any EIP!

Definition 1.20. A set of vectors $b_1, b_2, \dots, b_m$ is **orthonormal** w.r.t. an EIP if they are **orthogonal** ($\langle b_i, b_j \rangle = 0$ for $i \neq j$) and **unit** ($\langle b_i, b_i \rangle = 1$).

Theorem 1.21. Every linear subspace $L \subseteq \mathbb{R}^n$ has an orthonormal basis.

Gram-Schmidt Orthogonalization Process is an algorithm for converting an arbitrary basis $b_1, b_2, \dots, b_m$ of L into an orthonormal basis $f_1, f_2, \dots, f_m$ of L . For each t , let f_t = normalized $b_t - \sum_{j=1}^{t-1} \langle b_t, f_j \rangle f_j$. Each f_t is orthogonal to the previous f_j and has unit size. At the end we have m LI vectors $f_1, f_2, \dots, f_m$ in L , so they form a basis.

Definition 1.22. The **orthogonal complement** $L^\perp \subseteq \mathbb{R}^n$ of the linear subspace $L \subseteq \mathbb{R}^n$ is the set of vectors **orthogonal** to L , i.e. $L^\perp = \{y \in \mathbb{R}^n : \langle x, y \rangle = 0 \ \forall x \in L\}$.

Definition 1.23. The **orthogonal decomposition** of x w.r.t. $L, L^\perp$ is $x = x_L + x_{L^\perp}$.

Computing Orthogonal Decomposition: Start with x and an arbitrary basis of L . Compute an orthonormal basis of L using Gram-Schmidt. Compute $x_L = \sum_i \langle u_i, x \rangle u_i$ as the sum of 1D projections of x onto the orthonormal basis vectors u_i . Compute $x_{L^\perp} = x - x_L$.

1.4 Affine Subspaces

Definition 1.24. An **affine subspace** $M \subseteq \mathbb{R}^n$ is a set of the form $M = a + L$ where $a \in \mathbb{R}^n$ and $L \subseteq \mathbb{R}^n$ is a linear subspace. It can be represented as $M = \{x \in \mathbb{R}^n : Ax = b\}$.

Lemma 1.25. For an affine subspace M , $L = \{x - y : x, y \in M\}$ is the linear subspace.

This implies that the vector a can be any vector in M and you always get $M = a + L$.

Theorem 1.26. An affine subspace $M \subseteq \mathbb{R}^n$ is a nonempty set which is closed under **affine combinations** which are linear combinations with sum of coefficients = 1.

Definition 1.27. A **hyperplane** is an affine subspace of dimension $n - 1$.

2 Start to LP Duality

2.1 Basics

Definition 2.1. A **linear program** is a mathematical program with domain $= \mathbb{R}^n$, linear objective, and finite number of linear constraints. A common representation is

$$\min_{x \in \mathbb{R}^n} \{c^T x : Ax \geq b, x \geq 0\}$$

We can convert any LP into this form. Note that if x_1 is unconstrained in sign, we can replace $x_1 = y_2 - y_3$ where $y_2, y_3 \geq 0$.

Definition 2.2. A mathematical program is

- **feasible** if it has ≥ 1 feasible solution.
- **solvable** if it has ≥ 1 optimal solution.
- **bounded** if there is a bound on the objective of all feasible solutions.

Note solvable $\implies$ bounded but bounded $\not\implies$ solvable. For example, consider $\min_{x \in \mathbb{R}} \{x : x > 0\}$.

2.2 Polyhedral Representations

Definition 2.3. A **polyhedral representation** of a set $X \subseteq \mathbb{R}^n$ is a representation of the form $X = \text{Proj}_X(X^+)$ where X^+ is a higher-dimensional polyhedron. Equivalently $X = \{x \in \mathbb{R}^n : \exists w \in \mathbb{R}^k \text{ s.t. } Px + Qw \leq r\}$.

For example, consider $X = \{x \in \mathbb{R}^n : \|x\|_1 \leq 1\}$. This requires 2^n linear constraints. We could instead write $X^+ = \{x, w \in \mathbb{R}^n : w_i \geq x_i, w_i \geq -x_i, \sum w_i \leq 1\}$ and $X = \text{Proj}_X(X^+)$. X^+ only has $2n + 1$ linear constraints.

Fourier-Motzkin Elimination is a simple (but inefficient) LP solving algorithm. It involves repeatedly eliminating variables. Given $\text{OPT}(P) = \max_{x \in \mathbb{R}^n} \{c^T x : Ax \leq b, x \geq 0\}$, rewrite as $\text{OPT}(P) = \max_{(x,t) \in \mathbb{R}^{n+1}} \{t : Ax \leq b, t \leq c^T x, x \geq 0\}$. Now eliminate each x variable. To eliminate x_i , rewrite all constraints as $x_i \geq \dots, x_i \leq \dots$, or keep same if x_i not involved. Let M^+, M^-, M_0 be these sets of constraints. Make a new system excluding x_i where you add $m^+ \leq m^-$ for **every pair** of constraints m^+, m^- (and add M_0).

Note: FM Elimination is generally helpful for proving statements about linear programs. For example, it proves that every polyhedrally representable set is a polyhedron. Just start with X^+ and eliminate variables until you get X . It also proves that for LPs, bounded $\iff$ solvable. At the end, the feasible region for t is a closed interval which is bounded below (for min) or above (for max), and in both cases there is a t^* which attains the optimum value.

Definition 2.4. For a function $f : X \rightarrow \mathbb{R}$, the **epigraph** of f is $\text{Epi}(f) = \{(x, t) \in \mathbb{R}^{n+1} : t \geq f(x), x \in X\}$.

Definition 2.5. $f : X \rightarrow \mathbb{R}$ is a **polyhedrally representable function** if $\text{Epi}(f)$ is polyhedrally representable.

Theorem 2.6. $f(x) = \max_{1 \leq i \leq I} \{a_i^T x + b_i\}$ for finite I is a p.r. function, and all p.r. functions can be written in this form.

Definition 2.7. An **affine mapping** $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a function where $f(x) = Ax + b$. The **affine image** of X is $f(X)$. The **inverse affine image** of Y is $f^{-1}(Y)$.

Theorem 2.8. Polyhedra have the following closure properties.

1. **Intersection** of finitely many polyhedra is a polyhedron.

2. **Direct product** of finitely many polyhedra is a polyhedron.
3. **Affine image** of a polyhedron is a polyhedron.
4. **Inverse affine image** of a polyhedron is a polyhedron.
5. **Minkowski Sum** of polyhedra is a polyhedron.

and p.r. functions have the following **superposition theorem**: let $f_1, f_2, \dots, f_m, F : \mathbb{R}^n \rightarrow \mathbb{R}$ be p.r. where F is nondecreasing w.r.t. each variable. Then $g(x) = F(f_1(x), \dots, f_m(x))$ is a p.r. function.

2.3 Linear, Affine, Conic, Convex

Lemma 2.9. Any closed convex set $C \subseteq \mathbb{R}^n$ is the intersection of all closed half-spaces $s_i^T x \leq r_i$ containing C .

Definition 2.10.

1. **Linear combos** are $\sum_i \lambda_i x_i$
2. **Affine combos** are $\sum_i \lambda_i x_i$ with $\sum_i \lambda_i = 1$
3. **Conic combos** are $\sum_i \lambda_i x_i$ with $\lambda_i \geq 0$
4. **Convex combos** are $\sum_i \lambda_i x_i$ with $\sum_i \lambda_i = 1$ and $\lambda_i \geq 0$.

Definition 2.11. A **conic** set $K \subseteq \mathbb{R}^n$ is a nonempty set s.t. $\forall x \in K$ and $t \geq 0$ we have $tx \in K$. A **cone** is a conic and convex set.

Definition 2.12. A **LACC subspace** is a nonempty set which is closed under LACC combinations. The **LACC span / hull** of a set S is ...

1. **Inner**: The set of all LACC combinations of S .
2. **Outer**: The intersection of all LACC subspaces containing S .

Definition 2.13. For a set $S \subseteq \mathbb{R}^n$, the **dimension** of S is the dimension of $\text{aff}(S)$ (which is the dimension of the underlying linear subspace).

2.4 Classical Theorems

Theorem 2.14 (Carathéodory's Theorem). Let $X \subseteq \mathbb{R}^n$. All $x \in \text{conv}(X)$ can be represented as a convex combination of $\leq \dim(X) + 1$ points in X .

Proof Sketch. We will just focus on the $\dim(X) = n$ case. Let $x \in \text{conv}(X)$ be arbitrary. Write $x = \sum_{i=1}^k \lambda_i x_i$ where k is minimal. AFSOC $k > n + 1$. View

$$\sum_{i=1}^k \lambda_i x_i = x$$

$$\sum_{i=1}^k \lambda_i = 1$$

$$\lambda \geq 0$$

as a linear system in λ . The system is feasible and has k variables and $n+1$ equality constraints where $k > n+1$. Thus the solution set has dimension $k - (n+1) \geq 1$, so it intersects the boundary $\lambda \geq 0$, so there is a solution with some $\lambda_i = 0$. Contradiction. $\square$

Theorem 2.15 (Radon's Theorem). *Let $x_1, x_2, \dots, x_m \in \mathbb{R}^n$ be vectors with $m \geq n+2$. Then there exists a partition of $X = x_1, x_2, \dots, x_m$ into A, B s.t. $\text{conv}(A) \cap \text{conv}(B) \neq \emptyset$.*

Proof Strategy. We want to get some x' in both $\text{conv}(A)$ and $\text{conv}(B)$, so we want $x' = \sum_{i \in A} \gamma_i x_i = \sum_{j \in B} \delta_j x_j$ where $\sum_{i \in A} \gamma_i = 1$ and $\sum_{j \in B} \delta_j = 1$ and $\gamma \geq 0, \delta \geq 0$. We don't know how to split A and B , but we can get around that by replacing $\delta_j \leftarrow -\gamma_j$ and rewriting the constraints as $\sum_{i=1}^m \gamma_i x_i = 0$, $\sum_{i=1}^m \gamma_i = 0$, and γ unconstrained. There are m variables and $n+1$ equality constraints where $m > n+1$, so there is some $\bar{\gamma} \neq 0$ solution to the system. Based on our replacement of $\delta_j \leftarrow -\gamma_j$, we should let $A = \{x_i : \gamma_i > 0\}$ and $B = \{x_i : \gamma_i \leq 0\}$. One can verify that x' indeed works (with the final step being normalization to make sure $\sum_{i \in A} \gamma_i = \sum_{j \in B} \delta_j = 1$). $\square$

Theorem 2.16 (Helly's Theorem). *Let $A_1, A_2, \dots, A_m \subseteq \mathbb{R}^n$ be convex sets where $m \geq n+1$. If all collections of $n+1$ subsets share a common point, then all m subsets share a common point.*

Proof. Let's use induction on m . $m = n+1$ is trivial. Now assume the claim holds for $m-1$. We want to get some $x' \in A_1 \cap A_2 \cap \dots \cap A_m$. We know that there exists an $x^i \in A_1 \cap A_2 \cap \dots \cap A_{i-1} \cap A_{i+1} \cap \dots \cap A_m$ for each $1 \leq i \leq m$. By Radon's Theorem, we can split $x^1, x^2, \dots, x^m$ into A, B s.t. $\text{conv}(A) \cap \text{conv}(B) \neq \emptyset$. Reindex so that $A = \{x_1, x_2, \dots, x_{J-1}\}$ and $B = \{x_J, x_{J+1}, \dots, x_m\}$. A point in $\text{conv}(A) \cap \text{conv}(B)$ is in all m sets. $\square$

Theorem 2.17 (Separating Hyperplane Theorem). *Let $A, B \subseteq \mathbb{R}^n$ be disjoint nonempty convex sets. Then there exist $s, c \in \mathbb{R}^n$ with $s \neq 0$ s.t. $s^T x \leq c$ for all $x \in A$ and $s^T x \geq c$ for all $x \in B$. If A and B are closed, and A or B is bounded, then the separation can be made strict.*

Theorem 2.18 (Supporting Hyperplane Theorem). *Let $C \subseteq \mathbb{R}^n$ be a convex set and $x_0 \in \text{bd}(C)$. Then there exists a supporting hyperplane π at x_0 . Further, if $C \subseteq \mathbb{R}^n$ is a set with nonempty interior s.t. for all $x_0 \in \text{bd}(C)$ there is a supporting hyperplane π at x_0 , then C is convex – and C is the intersection of all half-spaces generated by the supporting hyperplanes.*

2.5 Farkas' Lemma and Certificates

Lemma 2.19 (Homogeneous Farkas' Lemma). $a^T x \geq 0$ is a consequence of $a_i^T x \geq 0$ for $i = 1, 2, \dots, m$ iff $a^T x \geq 0$ is a nonnegative linear combination of $a_i^T x \geq 0$ constraints.

Note that $\Leftarrow$ is easy. $\Rightarrow$ is interesting.

Theorem 2.20 (General Theorem on Alternatives). A linear system S (strict, nonstrict allowed) is **infeasible** iff a trivially infeasible inequality is a consequence of S .

Note there are two ways to get a trivially infeasible inequality by taking linear combinations of constraints in S – either some strict constraint is included, and you get $0^T x > \epsilon$ for $\epsilon \geq 0$, or no strict constraint is included, and you get $0^T x \geq \epsilon$ for $\epsilon > 0$.

Lemma 2.21 (Inhomogeneous Farkas' Lemma). $a^T x \geq b$ is a consequence of $a_i^T x \geq b_i$ for $i = 1, 2, \dots, m$ iff $a^T x \geq b$ is a nonnegative linear combination of $a_i^T x \geq b_i$ constraints and trivial $0 \leq 1$ constraints.

Let $X = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a polyhedron. How do we certify...

- **Consequence?** $c^T x \leq d$ is a consequence iff there exists a $\lambda \geq 0$ with $\lambda^T A = c$ and $\lambda^T b \leq d$.
- **Infeasibility?** Show $0^T x \geq \epsilon$ for $\epsilon > 0$ is a consequence of $Ax \leq b$.
- **Subset?** $Y = \{x \in \mathbb{R}^n : Cx \leq d\}$ has $X \subseteq Y$ iff $c_i^T x \leq d_i$ is a consequence of $Ax \leq b$ for all i .
- **Bounded?** Find $R > 0$ s.t. $x_i \leq R, -x_i \leq R$ are consequences of $Ax \leq b$ for all i .

Suppose we have an LP $= \max\{c^T x : Ax \leq b\}$. How do we certify...

- **Upper Bound?** Show $c^T x \leq d$ is a consequence of $Ax \leq b$ for some d . In fact, we only care about getting $\lambda^T A = c$ because then we get $d = \lambda^T b$.
- **Optimality?** Find a feasible x^* and show $c^T x \leq c^T x^*$ is a consequence of $Ax \leq b$.

2.6 LP Duality

Let $\text{OPT}(P) = \max_{x \in \mathbb{R}^n} \{c^T x : Ax \leq b\}$. A is an $m \times n$ matrix. All $\lambda \in \mathbb{R}^m$ with $\lambda \geq 0$ represent all consequence inequalities of $Ax \leq b$. We care about consequence inequalities where $\lambda^T A = c$ and we want to find the minimum $d = b^T \lambda$ s.t. $c^T x \leq d$ is a consequence inequality. The dual problem is $\text{OPT}(D) = \min_{\lambda \in \mathbb{R}^m} \{b^T \lambda : A^T \lambda = c, \lambda \geq 0\}$.

Theorem 2.22. Consider LP P and dual LP D . Then...

1. **Primal-Dual Symmetry:** Dual of D is P .
2. **Weak Duality:** $\text{OPT}(P) \leq \text{OPT}(D)$.
3. **Strong Duality:** $\text{OPT}(P) = \text{OPT}(D)$ if P is feasible and bounded.

Primal-Dual Symmetry is interesting. Note that Weak Duality is clear based on how we constructed D . Then Strong Duality is clear by PDS + Weak Duality.

Theorem 2.23. Consider LP P and dual LP D . Then one of the following holds:

1. Both are feasible and bounded.
2. One is feasible and unbounded, and the other is infeasible.
3. Both are infeasible.

Dual LP Derivations: If you want to construct the dual LP given the original LP has a different form, you can go through the consequence inequality derivation and use some extra logic. For example, suppose $\text{OPT}(P) = \max_{x \in \mathbb{R}^n} \{c^T x : Ax \leq b, x \geq 0\}$. Because $x \geq 0$, I just need $\lambda^T A \geq c$ and $\lambda \geq 0$ to get a consequence inequality with $c^T x$, because $c^T x \leq (\lambda^T A)x \leq \lambda^T b$. Thus $\text{OPT}(D) = \min_{\lambda \in \mathbb{R}^m} \{\lambda^T b : A^T \lambda \geq c, \lambda \geq 0\}$. For another example, suppose $\text{OPT}(P) = \max_{x \in \mathbb{R}^n} \{c^T x : Ax = b\}$. Now I just need $\lambda^T A = c$ and λ unconstrained, because the equality constraints will combine correctly regardless of the sign of the λ_i 's. Thus $\text{OPT}(D) = \min_{\lambda \in \mathbb{R}^m} \{\lambda^T b : A^T \lambda = c\}$.

Takeaway: The feasible region of $\text{OPT}(D)$ should describe all possible λ 's which can give consequence inequalities of the form $c^T x \leq d$.

Theorem 2.24 (Complementary Slackness). Let x, λ be feasible for P, D . Let $s_i = b_i - A_i^T x$ for $i = 1, 2, \dots, m$ be the primal slacks and $t_i = c_i - A_i^T \lambda$ for $i = 1, 2, \dots, n$ be the dual slacks. Then x, λ are optimal iff $\lambda_i s_i = 0$ for all $i = 1, 2, \dots, m$ and $x_i t_i = 0$ for all $i = 1, 2, \dots, n$.

Proof. We know $c^T x \leq \lambda^T A x \leq \lambda^T b$. x, λ are optimal iff we have equality. This holds iff $c^T x = \lambda^T A x$ and $\lambda^T A x = \lambda^T b$ giving $\lambda^T (b - A x) = 0$ and $x^T (c - A^T \lambda) = 0$. If $Ax \leq b$ is the constraint in P , then $\lambda \geq 0$ appears in D , so all components in $\lambda^T (b - Ax)$ are nonnegative, giving the CS condition. If $A^T \lambda \geq c$ is the constraint in D , then $x \geq 0$ appears in P , so all components in $x^T (c - A^T \lambda)$ are nonpositive, giving the CS condition. $\square$

Note: The CS Theorem holds regardless of whether P has $Ax \leq b$ or $Ax = b$ and regardless of $x \geq 0$ or x unconstrained.

3 Polyhedral Theory

3.1 Faces

Definition 3.1. Let $C \subseteq \mathbb{R}^n$ be a closed convex set. Then...

- A **face** $F \subseteq C$ is a nonempty convex subset where for all line segments $\overline{xy} \subseteq C$ which intersect F , we have $\overline{xy} \subseteq F$.
- A **supporting hyperplane** $\pi = \{x \in \mathbb{R}^n : c^T x = \delta\}$ at $\bar{x} \in \text{rbd}(C)$ is a hyperplane where $c^T x \leq \delta$ for all $x \in C$ and $c^T \bar{x} = \delta$.
- A **proper face** $F \subseteq C$ is a face where $F \neq C$.
- An **exposed face** $F \subseteq C$ is a face where $F = C$ or $F = \{x \in C : c^T x = \delta\}$ for some supporting hyperplane $c^T x = \delta$.

An example of where a face is **not** an exposed face is in a set $C \subseteq \mathbb{R}^2$ which is a rectangle adjacent to a quarter-circle. The top point of transition from rectangle to quarter-circle is a face but not an exposed face.

Fact 3.2. Let $X = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a polyhedron. Then...

- All faces $F \subseteq X$ are of the form $F = X \cap \{a_i^T x = b_i \mid \forall i \in I\}$ for some $I \subseteq [m]$.
- All faces are exposed faces.
- The faces of X form a laminar family.

3.2 Extreme Points

Definition 3.3. Let $C \subseteq \mathbb{R}^n$ be a closed convex set. Then an **extreme point** $\bar{x} \in C$ is a 0-dimensional face of C .

Theorem 3.4. Let $X = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a polyhedron. Then $v \in X$ is an **extreme point** of X if...

- **Geometric:** v is not the midpoint of a line segment in X .
- **Algebraic:** v has n L.I. tight constraints $a_i^T x = b_i \mid \forall i \in I$.

Note that the geometric characterization works for general convex sets. Further, the algebraic characterization comes from the description of faces in polyhedra.

Fact 3.5. Let $C \subseteq \mathbb{R}^n$ be a compact convex set. Then $C = \text{conv}(\text{extr}(C))$.

3.3 Cones

Recall that a **cone** is a conic and convex set.

Definition 3.6. A **polyhedral cone** $K \subseteq \mathbb{R}^n$ is a polyhedron and a cone.

Fact 3.7. $K \subseteq \mathbb{R}^n$ is a **polyhedral cone** iff it takes the form $K = \{x \in \mathbb{R}^n : Ax \leq 0\}$.

Definition 3.8. Let $C \subseteq \mathbb{R}^n$ be a closed convex set. Then...

- A **recessive direction** $d \in \mathbb{R}^n$ is a vector where C contains a ray starting at some $\bar{x} \in C$ with direction d .
- The **recessive cone** $\text{rec}(C) \subseteq \mathbb{R}^n$ is the collection of all recessive directions of C .
- The **lineality space** $\text{lineality}(C) \subseteq \mathbb{R}^n$ is $\text{lineality}(C) = \text{rec}(C) \cap \text{rec}(-C)$.

Fact 3.9. Let $C \subseteq \mathbb{R}^n$ be a closed convex set. Then...

- If d is a recessive direction from some $\bar{x} \in C$, it is a recessive direction from all $\bar{x} \in C$.
- The recessive cone is indeed a cone.
- The lineality space is indeed a linear subspace.
- C is bounded iff $\text{rec}(C) = \{0\}$.

Fact 3.10. Let $X = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a polyhedron. Then...

- $\text{rec}(X) = \{d \in \mathbb{R}^n : Ad \leq 0\}$.
- $\text{lineality}(X) = \{d \in \mathbb{R}^n : Ad = 0\}$ (this equals $\text{Null}(A)$)!

Definition 3.11. A **pointed cone** $K \subseteq \mathbb{R}^n$ is a cone with $\text{lineality}(K) = \{0\}$.

Fact 3.12. A cone $K \subseteq \mathbb{R}^n$ is pointed iff $0 \in \mathbb{R}^n$ is the only extreme point of K .

Definition 3.13. The **dual cone** $K^* \subseteq \mathbb{R}^n$ of a cone $K \subseteq \mathbb{R}^n$ is $K^* = \{y \in \mathbb{R}^n : y^T x \geq 0 \ \forall x \in K\}$.

Fact 3.14. If K is closed, then $K^{**} = K$.

Proof. $K^{**} = \{x \in \mathbb{R}^n : y^T x \geq 0 \ \forall y \in K^*\}$. Any half-space $y^T x \geq c$ which contains K has $c \leq 0$, because $0 \in K$. Further if $y^T x \geq c$ is a valid hyperplane for K where $c < 0$, then so is $y^T x \geq 0$. This is because if not – meaning some $x \in K$ has $y^T x < 0$ – then $tx, t \rightarrow \infty$ would have $y^T x \rightarrow -\infty$, a contradiction. Thus the intersection of all closed half-spaces containing K is exactly K^{**} , so $K^{**} = K$! $\square$

3.4 Extreme Rays

Definition 3.15. An **extreme ray** is a face of a pointed cone $K \subseteq \mathbb{R}^n$.

Note in the above definition we exclude rays where $d = 0$ (i.e. extreme points are excluded).

Theorem 3.16. Let $K = \{x \in \mathbb{R}^n : Ax \leq 0\}$ be a polyhedral cone. Then $r = td, d \neq 0$ is an **extreme ray** of X if...

- **Geometric:** If $x \in r$ and $x = y + z$ for $y, z \in K$, then $y, z \in r$.
- **Algebraic:** d has $n - 1$ L.I. tight constraints $a_i^T x = 0 \ \forall i \in I$.

Note that the geometric characterization works for general cones. Further, the algebraic characterization comes from the description of faces in polyhedra.

Fact 3.17. Let $K \subseteq \mathbb{R}^n$ be a pointed cone. Then $K = \text{cone}(\text{extray}(K))$.

3.5 Bases

Definition 3.18. A **base** $B \subseteq \mathbb{R}^n$ of a pointed cone $K \subseteq \mathbb{R}^n$ is a set of the form $B = \{x \in K : f^T x = 1\}$ that intersects every nontrivial ray of K .

Fact 3.19. Let $B \subseteq \mathbb{R}^n$ be a base of a pointed cone $K \subseteq \mathbb{R}^n$. Then...

- B is closed, convex, and bounded.
- $K = \{ty : y \in B, t \geq 0\}$.
- Extreme points of B are exactly the generators of extreme rays of K .

3.6 Interiors

Lemma 3.20. Let $S \subseteq \mathbb{R}^n$ be a convex set. Then $\text{int}(S) \neq \emptyset \iff \dim S = n$.

Proof. Suppose $\text{int}(S) \neq \emptyset$. Let $x \in \text{int}(S)$ and $B_\epsilon(x) \subseteq S$. Then x and $x + \epsilon e_i$ for $i = 1, 2, \dots, n$ are $n + 1$ affinely independent points in S , so $\dim S = n$.

Suppose $\dim S = n$. Let $x_1, x_2, \dots, x_{n+1}$ be $n + 1$ affinely independent points in S , and let $P = \text{conv}(x_1, x_2, \dots, x_{n+1}) \subseteq S$. Let $x' = \frac{1}{n+1} \sum x_i$ be the centroid of $x_1, x_2, \dots, x_{n+1}$. I claim $x' \in \text{int}(P)$ and thus $x' \in \text{int}(S)$.

To see why, note that P is a polytope, and thus its boundary is given by its facets. Suppose x' is on facet $F = P \cap \{a^i x = b^i\}$ where $a^i x \leq b^i$ is in the inequality description of P . But then $a^i x' = b^i$ and $a^i x_j \leq b^i$ for all $j = 1, 2, \dots, n + 1$ implies $a^i x_j = b^i$ for all $j = 1, 2, \dots, n + 1$. Thus $x_1, x_2, \dots, x_{n+1}$ lie on an affine subspace of dimension $\leq n - 1$, but they are $n + 1$ affinely independent points, a contradiction. Thus x' is **not** on any facet F , so x' has nonzero distance d to every facet F in P . Let $d' > 0$ be the minimum distance of x' to any facet F . Then $B_{d'}(x') \subseteq P$, so $x' \in \text{int}(P)$, so $x' \in \text{int}(S)$. Thus $\text{int}(S) \neq \emptyset$. $\square$

Lemma 3.20 is very useful for “interior”-type questions, because often it is impractical to find a small ball $B_\epsilon(x) \subseteq S$, but it is much more practical to show $\dim(S) = n$ by finding $n + 1$ affinely independent points in S .

If you have a polyhedron P and its inequality description $Ax \leq b$ for “interior”-type questions, and you want to show $x \in \text{int}(P)$, typically you want to show x satisfies every inequality strictly. Thus x has nonzero distance to every facet, so it has minimum nonzero distance d' to every facet, so $B_{d'}(x) \subseteq P$.

3.7 General Representation Theorem

Theorem 3.21 (General Representation Theorem). *Let $X = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a pointed polyhedron. Then $X = \text{conv}(\text{ext}(X)) + \text{rec}(X)$.*

This gives an inner description of polyhedra. If X contains lines then let $L = \text{lineality}(X)$ and suppose L has basis $f^1, f^2, \dots, f^k$. The above theorem can be extended by adding $f^1, -f^1, \dots, f^k, -f^k$ into the cone term. If X is a polytope, then $\text{rec}(X) = \{0\}$. If X is a pointed cone, then $\text{ext}(X) = \{0\}$.

4 Simplex Algorithm

Let $P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}$ and $\text{OPT}(P) = \min_{x \in P} c^T x$ where $b \in \mathbb{R}_+^m$ and $\text{rank}(A) = m$ be a standard form LP. Note A has m L.I. rows and n columns where $m \leq n$.

4.1 Algorithm

Find an initial basic feasible solution $\bar{x}$ with basis B . Find a k th basic direction which is improving. Move along the k th basic direction until a basic variable ℓ equals 0. Update the basis $B \leftarrow B \cup \{k\} - \{\ell\}$ and repeat.

4.2 Definitions

Definition 4.1.

- A **basis** $B \subseteq [n]$ is a subset of indices with $|B| = m$ and the columns of A corresponding to B are linearly independent.
- A **basic solution** $x \in \mathbb{R}^n$ is a solution where $x_j = 0$ for $j \notin B$, $A_B x_B = b$ for some basis $I_B \subseteq [n]$.
- A **basic feasible solution** $x \in \mathbb{R}^n$ is a basic solution which is also feasible.
- A **degenerate** basic solution $x \in \mathbb{R}^n$ has a basic variable equal to 0.
- A **feasible direction** $d \in \mathbb{R}^n$ at point $\bar{x} \in P$ is a vector where $\bar{x} + \theta d \in P$ for some $\theta > 0$.

- The ***k*th basic direction** $d \in \mathbb{R}^n$ of a basis $B \subseteq [n]$ is given by $d_k = 1, d_j = 0$ for $j \notin B \cup \{k\}$, and $d_B = -A_B^{-1}A_k$.
- The ***k*th reduced cost** of a basis $B \subseteq [n]$ is $c^T d = c_k - c_B^T A_B^{-1} A_k$ where d is the *k*th basic direction.

Fact 4.2. d is a feasible direction of P at point $\bar{x} \in P \iff Ad = 0$ and $d_i \geq 0$ for all i s.t. $x_i = 0$.

The *k*th reduced cost tells us whether the *k*th basic direction d is “improving.” We want $c^T(x + \theta d) < c^T x$ for the basic solution x which simplifies to $c^T d < 0$.

4.3 Correctness (Non-Degeneracy)

We now go through the algorithm and think about its correctness. We make the following crucial assumption: **our BFS's are always non-degenerate.** We will address this assumption in the next subsection.

Let's go through the algorithm. We will not explain how to get an initial basic feasible solution. It is easy to find a basic direction k which is improving by computing the reduced costs $\bar{c}$. Further, we can detect an optimal solution when $\bar{c} \geq 0$.

Theorem 4.3. Let $\bar{x}$ be a BFS with reduced costs $\bar{c}$. If $\bar{c} \geq 0$, then $\bar{x}$ is optimum, and if $\bar{x}$ is optimum and non-degenerate, then $\bar{c} \geq 0$.

When we move along the *k*th basic direction, this raises the following question.

Question 4.4. How do we know the *k*th basic direction is a feasible direction?

Answer 4.5. It's easy to check the *k*th basic direction has $Ad = 0$. Further, $x_i = 0$ only if i is a nonbasic variable (non-degeneracy), and $d_k = 1, d_j \geq 0 \ \forall j \notin B \cup \{k\}$, so $d_i \geq 0$ for all i s.t. $x_i = 0$.

Lastly, when we update the basis $B \leftarrow B \cup \{k\} - \{\ell\}$, this raises the following question.

Question 4.6. How do we know $B \cup \{k\} - \{\ell\}$ is a basis?

Answer 4.7. This can be shown directly, i.e. suppose you have a linear combination α of the columns of A corresponding to $B \cup \{k\} - \{\ell\}$ which equals 0, and you show that $\alpha = 0$.

Thus if our BFS's are always non-degenerate, the simplex algorithm terminates with an optimal solution in finite time.

4.4 Dealing with Degeneracy

In order to deal with degeneracy, we use **Bland's Rule**:

- Among the k th basic directions which are improving, pick the one with the smallest index.
- Among the basic variables ℓ which turn to 0, remove the one from B with the smallest index.

Degeneracy is still possible (for example, two basic variables could turn to 0 at the same time), but Bland's Rule guarantees the simplex algorithm terminates in finite time.

4.5 Extensions (Algebraic Sensitivity)

Suppose $\bar{x}$ is a BFS with basis B and is optimal to the LP.

Question 4.8. Suppose we slightly modify $c \leftarrow c + \delta e_i$. How can we recompute an optimal solution?

Answer 4.9. If $c \leftarrow c + \delta e_i$, then feasible region of LP does not change, but the objective values change. Viewing the reduced cost formula, if x_i is a nonbasic variable, we just need to check $\bar{c}_i \geq 0$. If x_i is a basic variable, we need to check $\bar{c} \geq 0$. If these are not true, we just need to run simplex algorithm for a few more steps to recompute an optimal solution.

Question 4.10. Suppose we slightly modify $b \leftarrow b + \delta e_j$ for $\delta > 0$. How can we recompute an optimal solution?

Answer 4.11. If $b \leftarrow b + \delta e_j$, then the objective values don't change, but the feasible region of LP changes. Dual simplex algorithm (not explained in class) will fix infeasibility.

Question 4.12. Suppose we want to add a new constraint $d^T x \leq d_0$ to the LP. How can we recompute an optimal solution?

Answer 4.13. If $\bar{x}$ is feasible for that constraint, it is still optimal to the LP. Otherwise, rewrite constraint as $d^T x + x_{n+1} = d_0$ and add new variable $x_{n+1} \geq 0$ into the LP. Dual simplex algorithm (not explained in class) will fix infeasibility.

Question 4.14. Suppose we want to add a new variable x_{n+1} to the LP. How can we recompute an optimal solution?

Answer 4.15. $\bar{x}$ is still feasible when setting $x_{n+1} = 0$. Thus we just need to check $\bar{c}_{n+1} \geq 0$. If that is not true, we just need to run simplex algorithm for a few more steps to recompute an optimal solution.

5 Final Topics

Note: Column generation appeared on my final exam, but Dantzig-Wolfe decomposition and Bender's decomposition were explicitly stated as not required.

5.1 Column Generation

Let $P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}$ and $\text{OPT}(P) = \min_{x \in P} c^T x$ where $b \in \mathbb{R}_+^m$ and $\text{rank}(A) = m$ be a standard form LP. Note A has m L.I. rows and n columns where $m \leq n$.

Suppose the number of variables n is very large. It is not tractable to find an optimal solution running simplex algorithm on the whole LP. Instead, we could start with the LP on a **subset** of the variables (and all other variables = 0), and strategically add variables into the LP, always maintaining an optimal solution. We call the LP we are maintaining the **master problem** and the IP we use to pick an adding variable the **pricing subproblem**.

Question 5.1. How should we build the initial master problem?

Answer 5.2. Find a BFS $\bar{x}$ with basis $B \subseteq [n]$ to the original LP. Then the initial master problem should contain only the basic variables. $\bar{x}$ is the only feasible solution to the initial master problem (m variables and m L.I. equality constraints).

Question 5.3. How should we pick the variable to add into the master problem?

Answer 5.4. We should add the variable with the lowest reduced cost because it is the most promising for improvement in $\bar{x}$. Now, we could just do this by iterating over the variables, computing reduced costs, and picking the lowest one, but the number of variables is very large, so this is slow.

Instead, we want a smarter method. Observe $\bar{c}_k = c_k - c_B^T (A_B)^{-1} A_k = c_k - p^T A_k$ for fixed vector $p = (A_B)^{-1T} c_B$. If we are lucky, all the c_k 's will be the same, like 1, and the A_k 's can be described by a feasible region IP of some integer program in a few variables. If so, we obtain the pricing subproblem where the objective is $1 - p^T a$ and $a \in IP$. This gives us the vector a and corresponding variable x_k with minimum reduced cost.

The Cutting Stock Problem is a great example of column generation.

5.2 Dantzig-Wolfe Decomposition

Let $P = \{x \in \mathbb{R}^n : Ax \leq a, Bx \leq b, x \geq 0\}$ and $\text{OPT}(P) = \min_{x \in P} c^T x$. Suppose the number of variables n is very large. Dantzig-Wolfe Decomposition is an algorithm for solving the LP efficiently. It involves a **reformulation** of the LP by using general representation theorem on $Bx \leq b$ and **column generation** on the resulting LP.

In particular, consider $P' = \{x \in \mathbb{R}^n : Bx \leq b, x \geq 0\}$. P' does not contain lines, so it is a pointed polyhedron. By the General Representation Theorem, we have $P' = \text{conv}(\text{ext}(P')) + \text{rec}(P')$. Thus all $x \in P'$ satisfy $x = \sum_i \lambda_i x^i + \sum_j \mu_j r^j$ for $\lambda, \mu \geq 0$ and $\sum_i \lambda_i = 1$. We could replace x with this representation in P and get rid of the $Bx \leq b$ constraints. This would reduce the number of constraints but increase the number of variables (because we now have a variable λ_i for each extreme point x^i and a variable μ_j for each

extreme ray r^j). To solve the resulting LP, we use column generation to slowly add λ_i and μ_j variables.

Typically this is useful when you have a subset of constraints $Ax \leq a$ over **all** variables and subsets of constraints $B_1x \leq b^1, B_2x \leq b^2, \dots, B_kx \leq b^k$ over **dis-joint subsets** of variables. This is because when you do decompositions of the $B_ix \leq b^i$ constraints and column generation, typically the pricing subproblem and total number of variables are much smaller.

5.3 Bender's Decomposition

Bender's Decomposition is a technique for solving LPs where an LP's constraints are split into a "master problem" and a "subproblem." The procedure starts by getting an optimal solution to the master problem and passing it to the subproblem. If it is feasible, we are done. If not, the subproblem passes a violated constraint, which is added into the master problem. This process repeats until a feasible solution to the subproblem is found.